

0 Self-Grading Declaration

Based on the guidelines provided, **I self-assign a grade of 2/2 for this homework.** I have made a legitimate effort to complete 100% of the problems.

1 Assignment 1

Assignment 1

4th order tensor representations.

- (a) Provide an expression for the components of $\mathbb{A} = \mathbf{1} \otimes \mathbf{1}$; i.e., give an expression for A_{ijkl} . In the above $\mathbf{1}$ is the second order identity tensor with components δ_{ij} .
- (b) Express $\mathbb{A}$ in Voigt form.
- (c) Express in Voigt form the 4th order symmetric identity $\mathbb{I}^{\text{sym}}$ whose components are

$$I_{ijkl} = \frac{1}{2}(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}). \quad (1)$$

Warning: the answer looks odd for an identity.

This is the first assignment in the course of CEE231: Mechanics of Solids that we are faced with 4th order Tensors, which serves as linear transformation for mapping second order tensors to second order tensors, similarly as how second order tensors are linear mappings of vectors.

The tensorial product of two tensors $\mathbf{B}$ and $\mathbf{C}$ is defined such that:

$$[\mathbf{B} \otimes \mathbf{C}]_{ijkl} = B_{ij}C_{kl}$$

Computing the product and remembering that $\mathbf{1}_{ij} = \delta_{ij}$:

$$\mathbb{A} = [\mathbf{1} \otimes \mathbf{1}]_{ijkl} = \mathbf{1}_{ij}\mathbf{1}_{kl} = \delta_{ij}\delta_{kl} \quad (2)$$

Where the product in **Eq. (2)** yields 1 when $i = j$ and $k = l$ and 0 otherwise. The Voigt form of the fourth-order-tensor is:

$$\mathbb{A}_{\text{Voigt}} = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Following a similar reasoning, the components of the 4th order symmetric identity, defined by **Eq. (1)**, are:

$$I_{ijkl} = \frac{1}{2}(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}) = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.50 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.50 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.50 \end{bmatrix}$$

2 Assignment 2

Assignment 2

Consider a material with a single reflective plane of symmetry whose normal is given by $\mathbf{e}_2$. Using symmetry considerations, show that C_{1312} must be equal to zero.

2.1 Preface

Previous homework assignments examined how to describe the motion and strain of bodies, without considering the causes that generated such responses. Conversely, homework assignment 4 focused exclusively on the stress tensor, addressing the resulting stress state in bodies without accounting for the corresponding deformations and motions arising from the initial configuration.

All these concepts, which were previously treated independently, become interconnected when energy considerations are introduced. The response of a body subjected to external actions is governed by two fundamental principles [1]:

- (i) the *Law of Balance of Energy*, and
- (ii) the *Law of imbalance of Entropy*.

These two principles are commonly known as the first and second laws of thermodynamics, respectively.

2.2 Constitutive Relations on Linear Elastic Materials

Linear elastic materials have the property that for every value of strain, there corresponds a unique value of stress. In other words, no energy is dissipated when the material is loaded, and the loading and unloading paths coincide.

In such a scenario, there exists a fourth-order tensor $\mathbb{C}$, called the *elastic modulus tensor*, such that the constitutive relation in **Eq. (3)** is satisfied:

$$\boldsymbol{\sigma} = \mathbb{C}\boldsymbol{\varepsilon} \quad (3)$$

Here, $\mathbb{C}$ is a linear mapping that turns the strain tensor $\boldsymbol{\varepsilon}$ into the stress tensor $\boldsymbol{\sigma}$. In indicial notation, this is expressed as:

$$\sigma_{ij} = \mathbb{C}_{ijkl} \varepsilon_{kl}$$

Since each index i, j, k, l can take three possible values (1, 2, 3), the tensor $\mathbb{C}$ has $3^4 = 81$ components in general. However, due to the minor and major symmetries of the stress and strain tensors, this number reduces to 21 independent components. Additional reductions occur for materials possessing specific symmetries, as will be demonstrated as follows.

2.3 Monoclinic Symmetry

Material symmetries arise from the local microstructure and are associated with invariance under specific rotations or reflections. For instance, a unidirectional fiber-reinforced composite exhibits a

plane of reflectional symmetry orthogonal to the fiber direction, and it is indeed a *Monoclinic material*[1]. Physically, this implies that the stress response to a given strain state remains unchanged under such a symmetry transformation. Mathematically, the components of the elasticity tensor $\mathbb{C}$ are invariant when expressed in two bases related by a material's symmetry transformation.

2.3.1 Material's Symmetry Transformation

For the monoclinic material over direction $\mathbf{e}_2$, the reference basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ switches to $\{\mathbf{e}_1^*, \mathbf{e}_2^*, \mathbf{e}_3^*\} = \{\mathbf{e}_1, -\mathbf{e}_2, \mathbf{e}_3\}$. Such that:

$$\mathbf{e}_1 = \mathbf{e}_1^*, \quad \mathbf{e}_2 = -\mathbf{e}_2^*, \quad \mathbf{e}_3 = \mathbf{e}_3^* \quad (4)$$

In order to aid the reasoning behind what this transformation does to the elasticity tensor, it is useful to express the components as showcased in **Eq. (5)**:

$$C_{ijkl} = \mathbb{C}(\mathbf{e}_i, \mathbf{e}_j, \mathbf{e}_k, \mathbf{e}_l) \stackrel{\text{def}}{=} (\mathbf{e}_i \otimes \mathbf{e}_j) : \mathbb{C}(\mathbf{e}_k \otimes \mathbf{e}_l), \quad (5)$$

Where $\mathbb{C}$ depends linearly on each of its four arguments ¹, or equivalently, it is a linear operation on the basis vectors. Symbolically:

$$\mathbb{C}(\mathbf{e}_i, \mathbf{e}_j, \mathbf{e}_k, \mathbf{e}_l) \quad (6)$$

2.3.2 Component C_{1312}

When trying to compute C_{1312}^* , in the rotated basis, the values $\mathbf{e}_i^* = \mathbf{e}_1^*$, $\mathbf{e}_j^* = \mathbf{e}_3^*$, $\mathbf{e}_k^* = \mathbf{e}_1^*$ and $\mathbf{e}_l^* = \mathbf{e}_2^*$ should be replaced in **Eq. (6)**. Remembering the symmetry conditions imposed in **Eq. (4)**, this turns into:

$$C_{1312} = -C_{1312}$$

Which only holds when $C_{1312} = 0$. Following a similarly reasoning, it is demonstrated that a monoclinic material exhibits 13 independent elastic constants, typically found in materials such as ZrO_2 and Ni-Ti alloys[1].

¹It can easily be seen that the "component operation" defined in **Eq. (3)** preserves the linearity of the operation by plugging in linear combinations of the nbasis vector values.

3 Assignment 3

Assignment 3

Consider a cube of single crystal albite and one of iron (see A&G 5.6[2] for the moduli of albite and Table 7.2 in the textbook[1] for those of iron) with side length l_0 . If the cubes are volumetrically compacted to cubes of smaller dimension $l = (1 - \Delta)l_0$, then the strain in the cubes will be $\boldsymbol{\varepsilon} = -\Delta \mathbf{1}$.

- (a) Express the strain in Voigt notation.
- (b) Compute the stress state in the cubes.
- (c) Do you observe anything unusual about the tractions on the surface of the cubes, viz., the loads that would have to be applied to the surface of the cube to cause such a deformation?

3.1 Voigt Strain Tensor

In order to use Voigt notation, the strain tensor $\boldsymbol{\varepsilon} = -\Delta \mathbf{1}$ is first expanded as:

$$\boldsymbol{\varepsilon} = \begin{bmatrix} -\Delta & 0 & 0 \\ 0 & -\Delta & 0 \\ 0 & 0 & -\Delta \end{bmatrix}$$

Using Voigt notation, the strain tensor is represented as a 6-component vector [1]:

$$\boldsymbol{\varepsilon}_{\text{Voigt}} = \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ 2\varepsilon_{23} \\ 2\varepsilon_{31} \\ 2\varepsilon_{12} \end{bmatrix} = -\Delta \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad (7)$$

3.2 Stress State

The elastic stiffness tensors for both materials, expressed in Voigt notation, are:

$$C_{\text{Voigt}}^{\text{Albite}} = \begin{bmatrix} 69.1 & 34.0 & 30.8 & 5.1 & -2.4 & -0.9 \\ 34.0 & 183.5 & 5.5 & -3.9 & -7.7 & -5.8 \\ 30.8 & 5.5 & 179.5 & -8.7 & 7.1 & -9.8 \\ 5.1 & -3.9 & -8.7 & 24.9 & -2.4 & -7.2 \\ -2.4 & -7.7 & 7.1 & -2.4 & 26.8 & 0.5 \\ -0.9 & -5.8 & -9.8 & -7.2 & 0.5 & 33.5 \end{bmatrix} \text{ GPa.}$$

$$C_{Voigt}^{Fe} = \begin{bmatrix} 231.4 & 134.7 & 134.7 & 0 & 0 & 0 \\ 134.7 & 231.4 & 134.7 & 0 & 0 & 0 \\ 134.7 & 134.7 & 231.4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 116.4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 116.4 & 0 \\ 0 & 0 & 0 & 0 & 0 & 116.4 \end{bmatrix} \text{ GPa.}$$

The stress vector can then be obtained from **Eq. (8)** using **Eq. (7)**:

$$\sigma_{Voigt} = C_{Voigt} \varepsilon_{Voigt} \quad (8)$$

Carrying out the multiplication in Python (see Section 4) yields:

$$\sigma_{Voigt}^{Albite} = \Delta \begin{bmatrix} -133.9 \\ -223.0 \\ -215.8 \\ 7.5 \\ 3.0 \\ 16.5 \end{bmatrix} \text{ GPa,} \quad \sigma_{Voigt}^{Fe} = \Delta \begin{bmatrix} -500.8 \\ -500.8 \\ -500.8 \\ 0.0 \\ 0.0 \\ 0.0 \end{bmatrix} \text{ GPa.}$$

3.3 Discussion

For the case of iron, the stress state is purely hydrostatic, as evidenced by the identical normal stresses and the absence of shear components. This is consistent with the symmetry of the cubic crystal, which presents a equal unit cell in all-3 princial directions.

In contrast, the albite crystal exhibits noticeable shear stresses and unequal normal stresses, indicating that the material does not deform uniformly under isotropic volumetric contraction. This anisotropic behavior reflects the lack of symmetry in its triclinic crystal structure.

Physically, this implies that in order to achieve the same uniform volume reduction, the applied surface tractions on an albite cube would need to vary across faces and possibly include tangential components—whereas for iron, only equal compressive pressures normal to the cube faces would be required.

4 Supporting Python Script

A supporting Python script was used to verify manual results and avoid performing cumbersome calculations by hand. The latest version can be found in [the GitHub repository](#).

Python 3.13 Code

```
import numpy as np

def run_assignment_3():
    print("Assignment 5 Results: Stress State\n")

    # Strain vector (Eq. 7)
    eps_voigt = -1 * np.array([[1], [1], [1], [0], [0], [0]]) # take      = 1

    # Elasticity matrices (in GPa)
    C_AlB = np.array([
        [69.1, 34.0, 30.8, 5.1, -2.4, -0.9],
        [34.0, 183.5, 5.5, -3.9, -7.7, -5.8],
        [30.8, 5.5, 179.5, -8.7, 7.1, -9.8],
        [5.1, -3.9, -8.7, 24.9, -2.4, -7.2],
        [-2.4, -7.7, 7.1, -2.4, 26.8, 0.5],
        [-0.9, -5.8, -9.8, -7.2, 0.5, 33.5]
    ])

    C_Fe = np.array([
        [231.4, 134.7, 134.7, 0, 0, 0],
        [134.7, 231.4, 134.7, 0, 0, 0],
        [134.7, 134.7, 231.4, 0, 0, 0],
        [0, 0, 0, 116.4, 0, 0],
        [0, 0, 0, 0, 116.4, 0],
        [0, 0, 0, 0, 0, 116.4]
    ])

    # Compute _voigt = C_voigt * _voigt
    sigma_AlB = C_AlB @ eps_voigt
    sigma_Fe = C_Fe @ eps_voigt

    print(" _voigt = \n", eps_voigt)
    print("\n _voigt^Albite (GPa) = \n", sigma_AlB)
    print("\n _voigt^Fe (GPa) = \n", sigma_Fe)

if __name__ == "__main__":
    run_assignment_3()
```

5 References

- [1] L. Anand and S. Govindjee, *Continuum Mechanics of Solids*. Oxford University Press, 07 2020.
- [2] L. Anand and S. Govindjee, *Example Problems for Continuum Mechanics of Solids*. Independently published, 2020. Publication date: July 25, 2020.